

MST Edge Cost

Time limit: 1.00 s

Memory limit: 512 MB

Given an undirected weighted graph, determine for each edge the minimum spanning tree cost if the edge must be included in the spanning tree.

Input

The first line has two integers n and m : the number of nodes and edges. The nodes are numbered $1, 2, \dots, n$.

The following m lines describe the edges. Each line has three integers a, b, w : there is an edge between nodes a and b with weight w .

You can assume that the graph is connected and simple and each edge appears at most once in the graph.

Output

For each edge in the input order, print the minimum spanning tree cost when the edge is included.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$
- $1 \leq w \leq 10^9$

Example

Input	Output
5 6 1 2 4 1 3 2 2 4 2 3 4 1 3 5 4 4 5 3	10 8 8 8 9 8